higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MECHANOTECHNICS N4

19 NOVEMBER 2019

This marking guideline consists of 7 pages.

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QUESTION 1

1.1

- Handling of workpieces✓
  Workers must use mechanised lifting apparatus to handle heavy work pieces during machining.✓

- Space around and above each machine✓
  There should be sufficient space around each machine so that the operator can work unhindered. There should also be sufficient space above each machine to allow free movement of overhead cranes.✓

- Transport routes and aisles✓
  Transport route and aisles should not be used to store materials and they should be kept open at all times for the safe transport of materials and workpieces. Transport routes and passages must be clearly marked and wide enough for the moving, handling and conveying of equipment.✓

- Carrying capacity of the floor✓
  Workshop floors have to be reinforced to safely carry the extremely heavy machinery. Floor areas where heavy cranes are used must be sufficiently reinforced to carry the extra load.✓

- Storage facilities✓
  Suitable storage facilities in the form of containers for workpieces before and after machining must be placed in the convenient areas near the machine so that the operator can reach them when placing workpieces into them or taking them out.✓

- Supply of services✓
  Supply routes and service points for electricity, water, gas, steam and compressed air must be within easy reach and should not obstruct the hoisting and conveying of workpieces. Supply routes and service points should preferably be housed in ducts in the floor or walls and should be protected by steel covers.✓ (6 × 2) (12)

1.2

- Change in mass of the test piece
- Change in the appearance of the piece's surface
- Changes in structure
- Changes in the mechanical properties
- Depth of corrosion attack
- Type and nature of corrosion
- Changes in the corrosion medium (Any 5 × 1) (5)

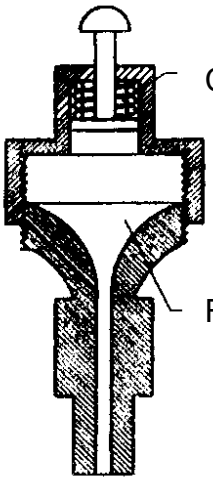
1.3

- Sandblasting cleaning
- Descaling
- Grease removal (3)

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1.4
- Applying force to the wrong ring during mounting
- Misalignment during mounting
- Incorrect mounting of housing end cover
- Mounting bearing in a damaged housing or shaft
- Bearing mounted the wrong way round
- Excessive lubrication (Any 5 × 1) (5)

1.5



(5)
[30]

QUESTION 2

$$\frac{T_1}{T_2} = e^{\mu\theta}$$

$$\frac{T_1}{T_2} = e^{0,3 \times 3,229} \checkmark$$

$$T_1 = 2,635T_2 \checkmark$$

$$T_2 = \frac{35,5 \times 10^3}{2,635} \checkmark$$

$$= 13472,486\ N \checkmark$$

$$P_t = (T_1 - T_2)v$$

$$= (35500 - 13472,486) \times 1,583 \checkmark$$

$$= 34869,555\ W \checkmark$$

$$P_f = F_f xv$$

$$= 2\ 500 \times 1,583 \checkmark$$

$$= 3957,5\ W \checkmark$$

$$P_g = P_t - P_f$$

$$= 34869,555 - 3957,5 \checkmark$$

$$= 30912,055\ W \checkmark$$

$$P_g = mgh$$

[14]

$$m = \frac{P_g}{g \times h}$$
$$m = \frac{30912,055}{9,81 \times 35} \checkmark\checkmark$$
$$= 90,031\ kg/s \checkmark$$
$$= 324,112\ t/h \checkmark$$

QUESTION 3

3.1 3.1.1

$$\mu = \frac{F_\mu}{N_R}$$
$$F_\mu = \mu \times N_R$$
$$F_\mu = \mu \times m \times g$$
$$F_\mu = 0,08 \times 200 \times 9,8 \checkmark\checkmark$$
$$= 156,96\ N \checkmark$$ (3)

3.1.2

$$F_\mu = 156,96 \times 0,75 \checkmark$$
$$F_\mu = 117,72\ N \checkmark$$
$$\mu = \frac{F_\mu}{m \times g}$$
$$\mu = \frac{117,72}{200 \times 9,81} \checkmark$$
$$= 0,06 \checkmark$$ (4)

3.2 3.2.1

$$P_i = \frac{2 \times \pi \times N \times T}{60}$$
$$mm\,T_{motor} = \frac{P_i \times 60}{2\pi N}$$
$$= \frac{3,5 \times 10^3 \times 60}{2 \times \pi \times 1\,750} \checkmark$$
$$= 19,099\ N.m \checkmark$$ (2)

3.2.2

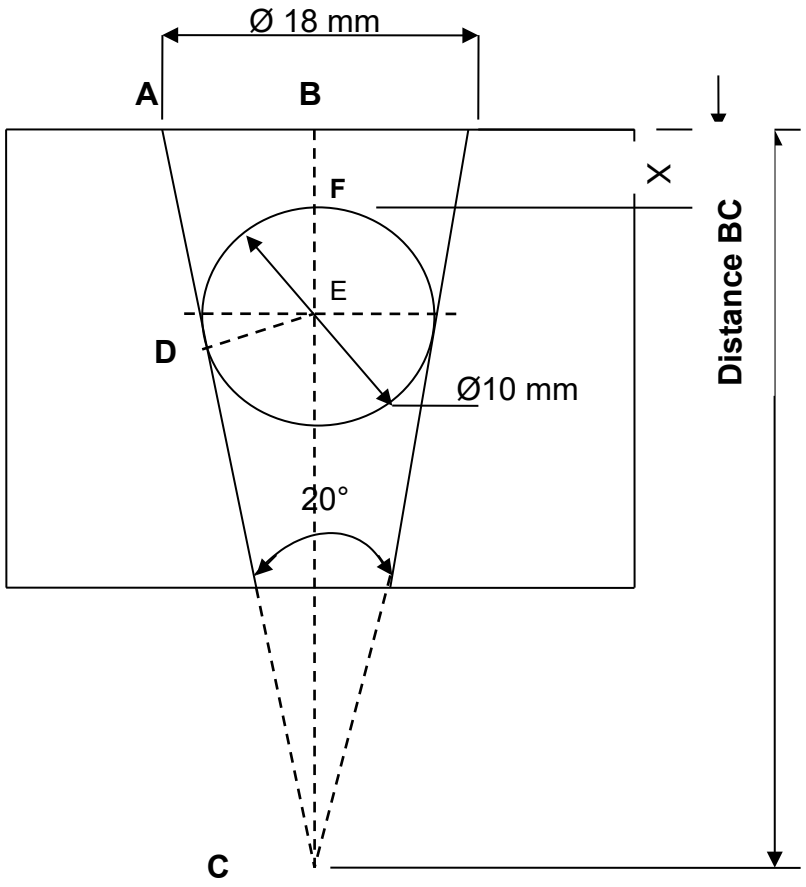
$$\eta = \frac{P_o}{P_i}$$
$$P_o = 0,8 \times 3,5 \times 10^3$$
$$= 2\,800\ W \checkmark$$
$$T_{spindle} = \frac{P_o \times 60}{2\pi N}$$
$$= \frac{2\,800 \times 60}{2 \times \pi \times 3\,500} \checkmark$$
$$= 7,639\ N.m \checkmark$$ (3)

3.2.3

$$T_{spindle} = \frac{P_o \times 60}{2\pi N}$$
$$= \frac{2\,800 \times 60}{2 \times \pi \times 25} \checkmark$$
$$= 1069,521\ N.m \checkmark$$ (2)

[14]

QUESTION 4



$X = BC - CE - EF$

Find BC

In ΔABC ✓

$\tan 10° = \frac{AB}{BC}$ ✓

$BC = \frac{9}{\tan 10°}$ ✓✓

$= 51{,}042 \ mm$ ✓

Find CE

In ΔCDE ✓

$\sin 10° = \frac{DE}{CE}$ ✓

$CE = \frac{5}{\sin 10°}$ ✓✓

$= 28{,}794 \ mm$ ✓

$X = BC - CE - EF$ ✓

$X = 51{,}042 - 28{,}794 - 5$ ✓✓

$= 17{,}288 \ mm$ ✓ **[14]**

QUESTION 5

5.1 (8)

$$T_P \times N_P = T_Q \times N_Q$$

$$N_Q = \frac{T_P \times N_P}{T_Q}$$

$$N_Q = \frac{60 \times 900}{140} \checkmark\checkmark$$

$$= 385{,}714\ r/\min \checkmark$$

but $N_Q = N_R = 385{,}714\ r/\min$ *because both gears are connected to the same shaft* ✓✓

$$T_R \times N_R = T_S \times N_S$$

$$N_S = \frac{T_R \times N_R}{T_S}$$

$$N_S = \frac{70 \times 385{,}714}{110} \checkmark\checkmark$$

$$= 245{,}454\ r/\min \checkmark$$

5.2 (6)

$$X = \frac{m}{2}(T_P + T_Q)$$

$$= \frac{3{,}5}{2}(60 + 140) \checkmark\checkmark$$

$$= 350\ mm \checkmark$$

$$Y = \frac{m}{2}(T_R + T_S)$$

$$= \frac{3{,}5}{2}(70 + 110) \checkmark\checkmark$$

$$= 315\ mm \checkmark$$

[14]

QUESTION 6

$$Q_1 = Q_2$$
$$v_1 \times A_1 = v_2 \times A_2$$
$$v_1 = \frac{v_2 \times A_2}{A_1}$$
$$v_1 = \frac{v_2 \times (0{,}135)^2}{(0{,}11)^2} \checkmark\checkmark$$
$$v_1 = 1{,}506 v_2 \checkmark$$

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + h_1 = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + h_2$$

$$\frac{160 \times 10^3}{10^3 \times 9{,}81} + \frac{v_1^2}{2 \times 9{,}81} = \frac{135 \times 10^3}{10^3 \times 9{,}81} + \frac{(1{,}506 v_1)^2}{2 \times 9{,}81} \checkmark\checkmark$$

$$16{,}310 + 0{,}051 v_1^{\ 2} = 13{,}762 + 0{,}116 v_1^{\ 2} \checkmark\checkmark$$
$$0{,}065 v_1^2 = 2{,}548 \checkmark$$
$$v_1^{\ 2} = 39{,}449 \checkmark$$
$$v_1 = 6{,}281\ m/s \checkmark$$

$$Q = v_1 \times A_1$$
$$Q = 6{,}281 \times \frac{\pi (0{,}135)^2}{4} \checkmark\checkmark$$
$$= 0{,}0899\ m^3/s \checkmark$$
$$= 89{,}90\ \ell/s \checkmark$$

[14]

TOTAL: 100

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